

Paper 2 - Set 07 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	E	Graphs of functions
2	A	Integration
3	B	Sequences and series
4	A	Identifying errors in proofs
5	B	Polynomials, factor and remainder theorems
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Worked solutions

Question 1 GRA-12 · Graphs of functions · spec MM8.1

Which of the following functions is undefined at exactly two real values of x ?

Answer: E

$x^2 - 4 = 0$ at $x = \pm 2$: two values. $x^2 + 4$ is never zero; $\frac{x}{x^2}$ and $\frac{1}{(x-2)^2}$ are undefined at one value; $\sqrt{x-2}$ is undefined for all $x < 2$.

Full method - how and why

'Undefined' means division by zero (or a negative under a square root). Count the offending values for each option.

- $\frac{1}{x^2 - 4}$: the denominator vanishes at $x = 2$ and $x = -2$ - **two** values.
why: Difference of two squares gives two distinct roots.
- $\frac{1}{x^2 + 4}$: the denominator is always at least 4, so it is never zero - defined everywhere.
why: A sum of a square and a positive constant is never zero.
- $\frac{1}{x}$ and $\frac{1}{(x-2)^2}$: each has a single problem value (0 and 2 respectively).
why: A repeated factor still vanishes at only one point.
- $\sqrt{x-2}$: undefined for **all** $x < 2$ - infinitely many values, not two.
why: Square roots need a non-negative argument.

Where it goes wrong

- Counting $(x-2)^2$ as two values because of the power.
- Thinking $x^2 + 4$ has roots ± 2 .

Takeaway Count the distinct values that break the expression: repeated factors give one, sums of squares give none.

Question 2 INT-10 · Integration · spec MM7.2

Find $\int \left(x + \frac{1}{x}\right)^2 dx$.

Answer: A

$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + x^{-2}$, which integrates to $\frac{x^3}{3} + 2x - x^{-1} + c$. (Option E is a common error: you cannot integrate a power of a bracket that way.)

Full method - how and why

Expand before integrating - there is no rule for integrating a power of a bracket on this specification.

- Expand: $\left(x + \frac{1}{x}\right)^2 = x^2 + 2\left(x \cdot \frac{1}{x}\right) + \frac{1}{x^2} = x^2 + 2 + x^{-2}$.
why: The cross term is $2 \times x \times \frac{1}{x} = 2$, a constant - easy to lose.
- Integrate term by term: $\frac{x^3}{3} + 2x + \frac{x^{-1}}{-1} = \frac{x^3}{3} + 2x - \frac{1}{x}$.
why: $\int x^{-2} dx = \frac{x^{-1}}{-1} = -\frac{1}{x}$; the index -2 is allowed because it is not -1 .
- Add the constant: $\frac{x^3}{3} + 2x - \frac{1}{x} + c$.
why: Indefinite integrals always need $+c$.

Where it goes wrong

- Choosing $\frac{1}{3}\left(x + \frac{1}{x}\right)^3 + c$ - you cannot integrate a bracket by raising its power.
- Sign error giving $+\frac{1}{x}$, or dropping the $2x$ from the cross term.

Takeaway Expand brackets fully first; $\int x^{-2} dx = -x^{-1}$, and the cross term of a square is easy to forget.

Question 3 SEQ-01 · Sequences and series · spec MM2.2

An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.

Answer: B

$$S_{40} = \frac{40}{2}(2 \cdot 5 + 39 \cdot 3) = 20(10 + 117) = 2540.$$

Full method - how and why

A direct application of the arithmetic sum formula. The only decision is which version of the formula to use.

- 1 Identify $a = 5$, $d = 3$, $n = 40$.

why: 'First term' and 'common difference' name a and d exactly; n is how many terms you are adding.

- 2 Use $S_n = \frac{n}{2}[2a + (n - 1)d]$: $S_{40} = 20[10 + 39 \times 3]$.

why: Use this version when you know a and d ; the other version $\frac{n}{2}(a + l)$ needs the last term first.

- 3 Evaluate the bracket: $10 + 117 = 127$, so $S_{40} = 20 \times 127 = 2540$.

why: Note $(n - 1)d = 39 \times 3$, not 40×3 - there are 39 steps between 40 terms.

Where it goes wrong

- Using $40d$ instead of $39d$, giving 2600.
- Forgetting to halve n , or halving the bracket instead.

Takeaway $S_n = \frac{n}{2}[2a + (n - 1)d]$ - the number of **gaps** is $n - 1$, one fewer than the number of terms.

Question 4 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Full method - how and why

Dividing an equation by a variable destroys the root where that variable is zero.

- 1 Line I divides $x^2 = 4x$ by x . That is only legitimate when $x \neq 0$ - and $x = 0$ does satisfy the equation. **First error.**

why: $0^2 = 4 \times 0$ ✓, so $x = 0$ is a genuine solution that the division removes.

- 2 The correct method: bring everything to one side and factorise. $x^2 - 4x = 0$, so $x(x - 4) = 0$.

why: Factorising preserves every root, because a product is zero exactly when a factor is.

- 3 So $x = 0$ or $x = 4$ - two solutions.

why: Line II's 'only solution' is wrong as a consequence of line I.

Where it goes wrong

- Blaming line II - it faithfully reports what line I produced; the first error is the division.
- Thinking $x^2 \div x = 2x$; it is x .
- Trying to 'fix' it by remembering to add $x = 0$ afterwards - correct in this case, but factorising is the reliable habit.

Takeaway Never divide an equation by a variable expression. Factorise instead: division loses roots, squaring adds them.

Question 5 X2-08 · Polynomials, factor and remainder theorems · spec MM1.6

$p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?

Answer: B

If $(x - 2)^2$ divides p then certainly $p(2) = 0$, so P is necessary for Q . But $p(x) = x - 2$ has $p(2) = 0$ without a repeated factor, so P is not sufficient.

Full method - how and why

The factor theorem gives one direction; a repeated factor is strictly more than a root.

- 1 If $(x - 2)^2$ is a factor, then $(x - 2)$ is a factor, so $p(2) = 0$: $Q \Rightarrow P$. **P is necessary.**

why: Factor theorem: $(x - a)$ is a factor exactly when $p(a) = 0$.

- 2 Is P sufficient? Take $p(x) = x - 2$: $p(2) = 0$, but $(x - 2)^2$ does not divide a degree-1 polynomial.

why: A single root is not a repeated root.

- 3 So P is necessary but not sufficient.

why: Note the direction: the weaker condition is necessary for the stronger one.

Where it goes wrong

- Answering 'sufficient but not necessary' - the common pattern, but reversed here.
- Thinking $p(2) = 0$ and $p'(2) = 0$ together would be needed; that is exactly the extra condition for a repeated root, and its absence is the point.

Takeaway $p(a) = 0 \Leftrightarrow (x - a)$ is a factor; a **repeated** factor also needs $p'(a) = 0$.

Question 6 RSN-14 · Chains of reasoning · spec Prf5

In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?

Answer: D

$P(M \text{ or } P) = 1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 0.3$.

Full method - how and why

The addition rule, with 'every student studies at least one' supplying the missing total.

- 1 'At least one of Maths and Physics' means $P(M \text{ or } P) = 1$.

why: Nobody is outside the union, so the union has probability 1.

- 2 Apply the addition rule: $P(M \cup P) = P(M) + P(P) - P(M \cap P)$.

why: Subtract the overlap, which the two percentages each count.

- 3 Substitute: $1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 1.3 - 1 = 0.3$.

why: The excess of 1.3 over 1 is exactly the double-counted overlap.

Where it goes wrong

- Multiplying $0.7 \times 0.6 = 0.42$ by assuming independence - nothing here says the subjects are independent, and the union condition rules it out.
- Answering 'cannot be determined'; the 'at least one' condition is precisely what makes it determinable.

Takeaway When percentages sum to more than 100% and everyone is covered, the excess **is** the overlap.

Question 7 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: **if n is a multiple of 6, then n is a multiple of 3**. Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Full method - how and why

The four relatives of an implication. Given $A \Rightarrow B$, you must be able to write the converse, inverse and contrapositive without hesitating.

- 1 Name the parts: A is ' n is a multiple of 6', B is ' n is a multiple of 3'.
why: Always label before manipulating - most errors here are bookkeeping errors.
- 2 The contrapositive is 'if not B then not A ': **if n is not a multiple of 3, then n is not a multiple of 6**.
why: Negate both parts **and** swap them. Doing only one of the two gives the wrong relative.
- 3 Sanity-check the meaning: if 3 does not divide n , then 6 certainly cannot, since every multiple of 6 is a multiple of 3.
why: The contrapositive is logically **equivalent** to the original, so it must also be true - a free check.

Where it goes wrong

- Choosing 'if n is a multiple of 3 then n is a multiple of 6' - that is the **converse** (swap only), and it is false ($n = 9$).
- Choosing 'if n is not a multiple of 6 then n is not a multiple of 3' - the **inverse** (negate only), also false ($n = 9$).

Takeaway Converse: swap. Inverse: negate. Contrapositive: swap **and** negate - and only the contrapositive is equivalent to the original.

Question 8 PRF-14 · Mathematical proof · spec Prf1

To prove '**if n^2 is even then n is even**' by contradiction, which assumption should be made?

Answer: B

Assume the hypothesis holds and the conclusion fails, then derive a contradiction: n odd gives n^2 odd, contradicting n^2 even.

Full method - how and why

Setting up a proof by contradiction for an implication: assume the hypothesis **and** the negated conclusion.

- 1 The claim is 'if n^2 is even then n is even': hypothesis n^2 even, conclusion n even.
why: Identify both parts before negating anything.
- 2 For contradiction, suppose the whole implication fails. An implication fails only when the hypothesis holds and the conclusion does not.
why: This is the one case that would make $A \Rightarrow B$ false.
- 3 So assume: n^2 is even **and** n is odd.
why: Both, not just one.
- 4 (The proof then runs: $n = 2k + 1 \Rightarrow n^2 = 4k^2 + 4k + 1$, odd - contradicting n^2 even.)
why: The contradiction is with the retained hypothesis.

Where it goes wrong

- Assuming only ' n is odd' and dropping the hypothesis - then there is nothing to contradict.
- Assuming ' n^2 is odd', which negates the hypothesis instead of the conclusion.
- Assuming ' n even and n^2 odd', which negates the wrong part of each.

Takeaway To contradict $A \Rightarrow B$, assume A **and** $\neg B$ - keep the hypothesis, negate only the conclusion.

Question 9 LGC-18 · The logic of arguments · spec Arg4

A statement reads: **for every positive integer n , there is a prime p with $n < p < 2n$** . Which of the following is its negation?

Answer: B

Negate the quantifiers in turn: 'for every n , there exists p with ...' becomes 'there exists n such that for every prime p , not ...', i.e. no prime lies strictly between n and $2n$.

Full method - how and why

Negating a nested $\forall\exists$ statement: work from the outside in, flipping one quantifier at a time.

1 Formalise: $\forall n \exists p$ (prime) with $n < p < 2n$.

why: Two quantifiers, so two flips.

2 Flip the outer: $\exists n$ such that $\neg[\exists p \text{ with } n < p < 2n]$.

why: 'For all' fails exactly when one case fails.

3 Flip the inner: $\exists n$ such that $\forall$ primes p , $\neg(n < p < 2n)$ - i.e. **no** prime lies strictly between n and $2n$.

why: 'There exists' fails exactly when every candidate fails.

4 So: there is a positive integer n for which no prime p satisfies $n < p < 2n$.

why: (The original is Bertrand's postulate and is in fact true, so this negation is false - but the question is about form.)

Where it goes wrong

- Keeping 'for every n ' and only negating the inside - that is a much stronger claim, not the negation.
- Turning it into ' $\exists n$ and $\exists p$ with $p \leq n$ or $p \geq 2n$ ', which is trivially true and says nothing: the inner quantifier must become **for all** primes.

Takeaway Negate nested quantifiers outside-in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, and the innermost statement is negated once.

Question 10 X2-28 · Statistics · spec M6.3

A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?

- I The largest value is at least 12.
- II The smallest value is at most 8.
- III The range is at least 4.

Answer: H

The two values above the median are distinct integers greater than 10, so at least 11 and 12 (I); similarly the two below are at most 9 and 8 (II); hence the range is at least $12 - 8 = 4$ (III).

Full method - how and why

Extremal reasoning with **distinct integers**: the distinctness forces gaps of at least 1 on each side of the median.

1 Order the five values. The median (third value) is 10.

why: Five values means the middle one is the third.

2 The two above are distinct integers greater than 10, so at least 11 and 12. Hence the largest is ≥ 12 . **I true.**

why: Distinct integers above 10 must be 11 or more, and two of them cannot both be 11.

3 Symmetrically, the two below are at most 9 and 8, so the smallest is ≤ 8 . **II true.**

why: Same argument downwards.

4 Range $\geq 12 - 8 = 4$. **III true.**

why: Combining the two bounds. (The mean being 10 is consistent - e.g. 8, 9, 10, 11, 12 - but is not needed for any of the three claims.)

5 So all three.

why: Match to the list.

Where it goes wrong

- Forgetting 'distinct' and allowing two values equal to 11, which would break I.
- Assuming the mean condition must be used; here it is extra information.

Takeaway Distinct integers force minimum gaps: k values above a median m reach at least $m + k$.

Question 11 RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?

Answer: E

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

Full method - how and why

Turn a two-variable Diophantine equation into a product by the 'add a constant and factorise' trick.

- 1 Set up: $ab = 2a + 2b$ with a, b positive integers.

why: Area equals perimeter, numerically.

- 2 Rearrange: $ab - 2a - 2b = 0$, then add 4 to both sides: $ab - 2a - 2b + 4 = 4$.

why: The added constant is the product of the two subtracted coefficients, chosen to make the left side factorise.

- 3 Factorise: $(a - 2)(b - 2) = 4$.

why: Check by expanding: $ab - 2a - 2b + 4$ ✓.

- 4 Positive factor pairs of 4 with $a - 2 \leq b - 2$: (1, 4) and (2, 2), giving 3×6 and 4×4 .

why: Negative factors would need $a < 2$; $a = 1$ gives $b - 2 = -4$, i.e. $b = -2$, not a length.

- 5 So **two** rectangles.

why: Check: 3×6 has area 18 and perimeter 18 ✓; 4×4 has 16 and 16 ✓.

Where it goes wrong

- Counting 3×6 and 6×3 separately - the question says to count each rectangle once.
- Forgetting to add 4 and being unable to factorise.
- Including the factor pair (4, 1) as a third solution.

Takeaway $ab + pa + qb = r$ factorises after adding pq : $(a + q)(b + p) = r + pq$. Then list factor pairs.

Question 12 X2-02 · Differentiation · spec MM6.3

Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?

Answer: C

If $f' > 0$ everywhere, f is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once: P is sufficient. But $f(x) = x^3$ has exactly one root and $f'(0) = 0$, so P is not necessary.

Full method - how and why

A necessary/sufficient question about polynomials. Test each direction, and look for a boundary case to break the converse.

- 1 Sufficiency: if $f'(x) > 0$ for all x , then f is strictly increasing, so it takes each value at most once - at most one root.

why: A strictly increasing function cannot return to a value it has already passed.

- 2 Moreover such an f must have odd degree (an even-degree polynomial turns), so it goes from $-\infty$ to $+\infty$ and crosses the axis. Exactly one root: $P \Rightarrow Q$. **Sufficient.**

why: Both halves are needed: 'at most one' plus 'at least one'.

- 3 Necessity: find an f with exactly one root but f' not always positive. Take $f(x) = x^3$: one root at 0, but $f'(0) = 0$.

why: A stationary point of inflexion satisfies Q without satisfying P .

- 4 So P is sufficient but not necessary.

why: Match to the list.

Where it goes wrong

- Claiming P is necessary because 'one root suggests increasing' - x^3 and also x^2 -type shifts break that.
- Forgetting that $f' > 0$ forces odd degree, so 'at least one root' still needs justifying.

Takeaway Strictly increasing $\Rightarrow$ at most one root; but a single root does not force a strictly positive derivative (x^3).

Question 13 BIN-12 · Binomial expansion · spec MM2.4

The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?

I $k = 0$

II $k = 3$

III $k = -6$

Answer: H

General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$ has power $9 - r - 2r = 9 - 3r$, which takes values 9, 6, 3, 0, -3, ..., -18. $k = 0$ needs $r = 3$: yes. $k = 3$ needs $r = 2$: yes. $k = -6$ needs $r = 5$: yes. All three.

Full method - how and why

Which powers of x actually occur. Find the general power as an expression in r and see which target values it can hit.

1 General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$, whose power of x is $(9 - r) + (-2r) = 9 - 3r$.

why: Add the powers contributed by each bracket; the $-\frac{1}{x^2}$ contributes -2 per copy.

2 As r runs from 0 to 9, the power takes the values 9, 6, 3, 0, -3, -6, -9, -12, -15, -18.

why: They step down by 3 - so only multiples of 3 appear, all the way down to -18.

3 $k = 0$ needs $r = 3$ ✓; $k = 3$ needs $r = 2$ ✓; $k = -6$ needs $r = 5$ ✓.

why: Each value of r must be a whole number between 0 and 9 - that is the only restriction.

4 All three occur: 'I, II and III'.

why: Solve $9 - 3r = k$ for each target and check that r is a valid index.

Where it goes wrong

- Forgetting that the second bracket contributes x^{-2r} , not x^{-r} .
- Assuming negative powers cannot appear because it is a 'binomial expansion'.

Takeaway Express the power of x as a linear function of r ; a term exists exactly when the required r is an integer in range.

Question 14 RSN-03 · Chains of reasoning · spec Prf5

Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?

Answer: D

If four people each shook hands with all four others, everyone shook hands with everyone, so the fifth person also made 4 handshakes, not 1. The others can happen: B is a pentagon of handshakes, C is everyone with everyone, D is a chain, E leaves one person out.

Full method - how and why

A handshake (graph) consistency question. Test each list for an internal contradiction.

- 1 Take 4, 4, 4, 4, 1: four people each shook hands with all four others.
why: With five people, the maximum possible is 4 handshakes each.
- 2 If person A shook hands with everyone, that includes the fifth person. The same for the other three.
why: Each of the four 'maximum' people contributes one handshake to the fifth person.
- 3 So the fifth person made at least 4 handshakes, not 1. **Impossible.**
why: The list contradicts itself.
- 4 The others are all realisable: 2, 2, 2, 2, 2 is a pentagon; 4, 4, 4, 4, 4 is everyone with everyone; 1, 1, 2, 2, 2 is a chain $A-B-C-D-E$; and 0 for one person just leaves them out.
why: Constructing one arrangement is enough to show a list is possible.

Where it goes wrong

- Rejecting 'everyone shakes four hands' as too many - it is exactly the complete case and is fine.
- Rejecting the list 1, 1, 2, 2, 2 without trying to build it; a chain works.
- Forgetting the parity check: the degree total must be even. $4 + 4 + 4 + 4 + 1 = 17$ is odd, which is a second, instant reason it is impossible.

Takeaway Two fast checks on a handshake list: the total must be even, and someone shaking everyone's hand forces everyone else's count up.

Question 15 LGC-06 · The logic of arguments · spec Arg1

A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?

Answer: A

'A only if B' means ' $A \Rightarrow B$ ': passing implies revising. It does not promise that revising guarantees a pass.

Full method - how and why

The phrase 'only if' is the one piece of logical English almost everyone mistranslates. 'A only if B' means $A \Rightarrow B$.

- 1 Let A = 'you pass', B = 'you revise'. The sentence is 'A only if B'.
why: Label first.
- 2 'A only if B' says that A can happen **only** in the case that B holds - so A forces B: $A \Rightarrow B$.
why: Compare with 'A if B', which is the other direction $B \Rightarrow A$. 'Only if' and 'if' point opposite ways.
- 3 Translate back: **if you pass, then you revised.**
why: Revising is necessary for passing, but the teacher promises no guarantee that it is enough.

Where it goes wrong

- Reading it as 'if you revise, you will pass' - the converse, and a much stronger promise than the teacher made.
- Reading it as 'if and only if'; 'only if' gives one direction only.
- Choosing 'if you do not pass then you did not revise' - the inverse, not equivalent (the correct contrapositive is 'if you did not revise, you did not pass').

Takeaway 'A if B' means $B \Rightarrow A$; 'A only if B' means $A \Rightarrow B$. 'Only if' marks the **necessary** condition.

Question 16 ERR-13 · Identifying errors in proofs · spec Err1

Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

I Suppose $x + \frac{1}{x} \geq 2$.

II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.

III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.

IV Hence the claim is proved.

Which of the following best describes the argument?

Answer: A

Starting from the conclusion and reaching a truth does not prove the conclusion unless every step is reversible and the argument is presented in the forward direction. Here the steps are reversible, so the repair is to write it backwards.

Full method - how and why

Assuming the conclusion. The chain of steps is valid but it runs the wrong way.

1 Line I supposes $x + \frac{1}{x} \geq 2$ - which is exactly the claim to be proved. **First error.**

why: A proof may suppose the **hypothesis** ($x > 0$), never the conclusion.

2 Lines II and III are then correct algebra, ending at $(x - 1)^2 \geq 0$, a truth.

why: Deriving a true statement from an assumption proves nothing: false statements can also imply true ones.

3 The repair: run it backwards. Start from $(x - 1)^2 \geq 0$ (known true), expand to $x^2 - 2x + 1 \geq 0$, so $x^2 + 1 \geq 2x$, and divide by $x > 0$ to get $x + \frac{1}{x} \geq 2$.

why: Every step is reversible here - the division by x is safe because $x > 0$ - so the reversal is valid.

4 So the best description is: it assumes what it is proving, and would be valid reversed.

why: Both halves matter: the flaw, and the fact that it is repairable.

Where it goes wrong

- Calling the argument correct because each line follows from the previous one - direction is what matters.
- Calling the claim false; it is true, as the reversed proof shows.
- Thinking multiplying by x reverses the inequality - $x > 0$ is given.

Takeaway Working backwards is how you **find** a proof; to **present** one, start from a known truth and check every step reverses.

Question 17 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Full method - how and why

Ordering Euclid's proof. Again the skeleton is assume-deduce-contradict, and the dependencies fix the order.

- 1 II first: assume finitely many primes $p_1, \dots, p_n$ and construct $N = p_1 p_2 \cdots p_n + 1$.
why: Everything else refers to N , so it must be defined first.
- 2 Then I: each p_i divides $p_1 \cdots p_n$, so dividing N by p_i leaves remainder 1 - no listed prime divides N .
why: This is the key property of the $+1$.
- 3 Then IV: $N > 1$, so it has at least one prime factor.
why: Every integer greater than 1 has a prime factor - a separate fact, independent of I.
- 4 Finally III: that prime factor is not in the list, contradicting the assumption that the list was complete. Order: II, I, IV, III.
why: The contradiction needs **both** I (none of the listed primes works) and IV (some prime does).

Where it goes wrong

- Putting III immediately after II, before either supporting fact has been established.
- Thinking the proof claims N is prime - it need not be; it merely has a prime factor outside the list.

Takeaway In a contradiction proof, the contradiction is the last line, and it must rest on everything before it.

Question 18 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I $p(x) = 0$ has at least one real root.
- II $p(x) = 0$ has at most four real roots.
- III The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.

Full method - how and why

Three claims about a quartic. Two follow from degree; the first needs a counterexample.

1 I: must a degree-4 polynomial have a real root? No: $x^4 + 1 \geq 1 > 0$ for all real x . **False.**
why: Even-degree polynomials can stay entirely above the axis - unlike odd-degree ones.

2 II: a polynomial of degree n has at most n real roots, so at most 4. **True.**
why: Each root supplies a linear factor; more than 4 would exceed the degree.

3 III: as $x \rightarrow \pm \infty$, a quartic tends to the same infinity at both ends (the sign of the leading coefficient).
why: Even powers dominate and are positive at both extremes.

4 A continuous function going to the same infinity both ways must turn round somewhere, so it has at least one turning point. **True.**
why: It has a global minimum (or maximum) at a finite point.

5 So II and III.
why: Match to the list.

Where it goes wrong

- Accepting I by analogy with odd-degree polynomials, which **must** have a real root.
- Reading II as 'exactly four roots' - repeated roots and non-real roots reduce the count.

Takeaway Odd degree guarantees a real root; even degree guarantees a turning point. Degree n caps the real roots at n .

Question 19 X2-13 · Binomial expansion · spec MM2.4

n is a positive integer. Which of the following statements is/are true?

- I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.
- II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.
- III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

Answer: H

I is the general term with $r = n$. II: put $x = 1$. III: expanding $(1+x)^n$ gives $1+nx$ plus non-negative terms when $x \geq 0$.

Full method - how and why

Three binomial facts: the general term, substituting a value, and a term-by-term inequality.

- 1 I: the coefficient of x^r in $(1+x)^m$ is $\binom{m}{r}$. With $m = 2n$ and $r = n$ that is $\binom{2n}{n}$. **True.**
why: Straight from the binomial theorem.
- 2 II: the sum of the coefficients of any polynomial is its value at $x = 1$. Here $(1-1)^n = 0^n = 0$ for $n \geq 1$. **True.**
why: Substituting $x = 1$ makes every power equal 1, so the value is the coefficient sum.
- 3 III: for $x \geq 0$, $(1+x)^n = 1+nx + \binom{n}{2}x^2 + \dots$, and every omitted term is non-negative. So $(1+x)^n \geq 1+nx$. **True.**
why: All binomial coefficients and all powers of a non-negative x are non-negative.
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Thinking II fails for $n = 0$ - the question says n is a positive integer.
- Doubting III for large n ; the extra terms only help, since $x \geq 0$.
- Confusing $\binom{2n}{n}$ with $\binom{2n}{2}$ or $2\binom{n}{2}$.

Takeaway Sum of coefficients = value at $x = 1$; and $(1+x)^n \geq 1+nx$ for $x \geq 0$ (Bernoulli) follows from discarding non-negative terms.

Question 20 LIN-10 · Straight lines · spec MM3.1

The line through $(a, 0)$ and $(0, b)$, with $a, b > 0$, passes through $(2, 3)$. Which of the following statements is/are true?

- I** $\frac{2}{a} + \frac{3}{b} = 1$
II $a > 2$ and $b > 3$
III The smallest possible value of ab is 24

Answer: H

The line is $\frac{x}{a} + \frac{y}{b} = 1$, so I holds. Since $\frac{2}{a} < 1$ and $\frac{3}{b} < 1$, II holds. $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$; setting $a = 2 + t$ gives $\frac{3(2+t)^2}{t} = 3(\frac{4}{t} + 4 + t) \geq 3(4 + 4) = 24$ by AM-GM (equality at $t = 2$, $a = 4$, $b = 6$). III holds.

Full method - how and why

The intercept form of a line, plus an optimisation. Statements I and II are quick; III needs a minimisation argument.

- I: a line with intercepts a and b has equation $\frac{x}{a} + \frac{y}{b} = 1$; substituting $(2, 3)$ gives $\frac{2}{a} + \frac{3}{b} = 1$. **True.****
why: The intercept form is worth knowing precisely for questions like this.
- II: since $\frac{2}{a}$ and $\frac{3}{b}$ are both positive and sum to 1, each is less than 1, so $a > 2$ and $b > 3$. **True.****
why: Two positive numbers summing to 1 are each below 1 - a clean argument with no algebra.
- III: from I, $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$. Put $a = 2 + t$ with $t > 0$: $ab = \frac{3(2+t)^2}{t} = 3\left(\frac{4}{t} + 4 + t\right)$.**
why: The substitution $a = 2 + t$ clears the awkward denominator and exposes the structure $\frac{4}{t} + 4 + t$.
- By AM-GM, $\frac{4}{t} + t \geq 2\sqrt{4} = 4$, with equality when $t = 2$. So $ab \geq 3(4 + 4) = 24$, attained at $a = 4$, $b = 6$. **True.****
why: $u + \frac{k}{u} \geq 2\sqrt{k}$ for positive u - here it gives both the bound and the case of equality, which is what 'smallest possible value' demands.
- All three are true.**
why: Match to the options.

Where it goes wrong

- Assuming II fails because a and b 'could be anything' - the positivity constraint forces both bounds.
- Claiming III without checking equality is attainable; a lower bound is only the minimum if some case achieves it.

Takeaway $\frac{x}{a} + \frac{y}{b} = 1$ is the intercept form; and $t + \frac{4}{t} \geq 4$ (AM-GM) is the standard tool for such minimisations.